Statistical Methods for Computer Science
Mock Intermediate Test 1 - 2025-26

First Name and Surname: ______________________

Student Registration Number: ______________________

1. The duration X (in months) for a shipment of car tyres used on a population of 900 cars is characterized by the following distribution:

Duration	$(0-20]$	$(20-40]$	$(40-60]$	$(60-80]$
Freq.	200	250	350	100

   Assuming equi-distribution with the classes:
   (a) Calculate the mean duration.
   (b) Calculate the median duration.
   (c) Calculate the standard deviation of duration.

   Answers:
   (a) [ ] / 1
   (b) [ ] / 2
   (c) [ ] / 2

2. The following table displays the joint probability distribution of two random variables X and Y.

		X		
		1	2	3
Y	1	0.10	0.08	0.06
	2	0.16	0.1	0.11
	3	0.02	0.16	0.21

   (a) Find the expected value of Y
   (b) Find $P[X = 1]$ in the case that $Y \geq 2$

   Answers:
   (a) [ ] / 2
   (b) [ ] / 3

3. The number of car accidents during weekends at a certain crossroad is a Poisson random variable with expected value equal to 0.7.
   (a) Find the probability of having at least three car accidents during a single weekend.
   (b) Consider the number of car accidents in different weekends as independent and identically distributed random variables. Find the probability that in this crossroad there are two car accidents in three weekends.

   Answers:
   (a) [ ] / 2
   (b) [ ] / 3

Parametric families for discrete random variables

Distribution	Support	$p_X(x)$	$\mathbb{E}[X]$	$\mathbb{V}[X]$
$X \sim \text{Bernoulli}(p)$ $p \in [0,1]$	$X \in \{0,1\}$	$p^x(1-p)^{1-x}$	p	$p \cdot (1-p)$
$X \sim \text{Binomial}(n,p)$ $n \in \{1,2,\ldots\}; p \in [0,1]$	$X \in \{0,1,\ldots,n\}$	$\binom{n}{x}p^x(1-p)^{n-x}$	$n \cdot p$	$n \cdot p \cdot (1-p)$
$X \sim \text{Poisson}(\lambda)$ $\lambda > 0$	$X \in \{0,1,\ldots,\}$	$e^{-\lambda}\dfrac{\lambda^x}{x!}$	λ	λ

Parametric families for continuous random variables

Distribution	Support	$f_X(x)$	$\mathbb{E}[X]$	$\mathbb{V}[X]$
$X \sim \text{Normal}(\mu,\sigma^2)$ $\mu \in \mathbb{R}; \sigma^2 > 0$	$X \in \mathbb{R}$	$\dfrac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\dfrac{1}{2\sigma^2}(x-\mu)^2\right\}$	μ	σ^2